Qualcomm Technologies, Inc.

GPR Packet Sniffer Tool

User Guide

80-VN500-12 Rev. AB

September 30, 2020

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Revision history

Revision	Date	Description
AA	May 2020	Initial release
AB	September 2020	Update for tool v2. Numerous changes were made to this document, it should be read in entirety.

Note: There is no Rev. I, O, Q, S, X, or Z per Mil. standards.

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1 Introduction

1.1 Purpose

This document describes how to use the GPR Packet Sniffer tool.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **`copy a:*.* b:`**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

Shading indicates content that has been added or changed in this revision of the document.

1.3 Technical assistance

For assistance or clarification on information in this document, submit a case to Qualcomm Technologies, Inc. (QTI) at https://createpoint.qti.qualcomm.com/.

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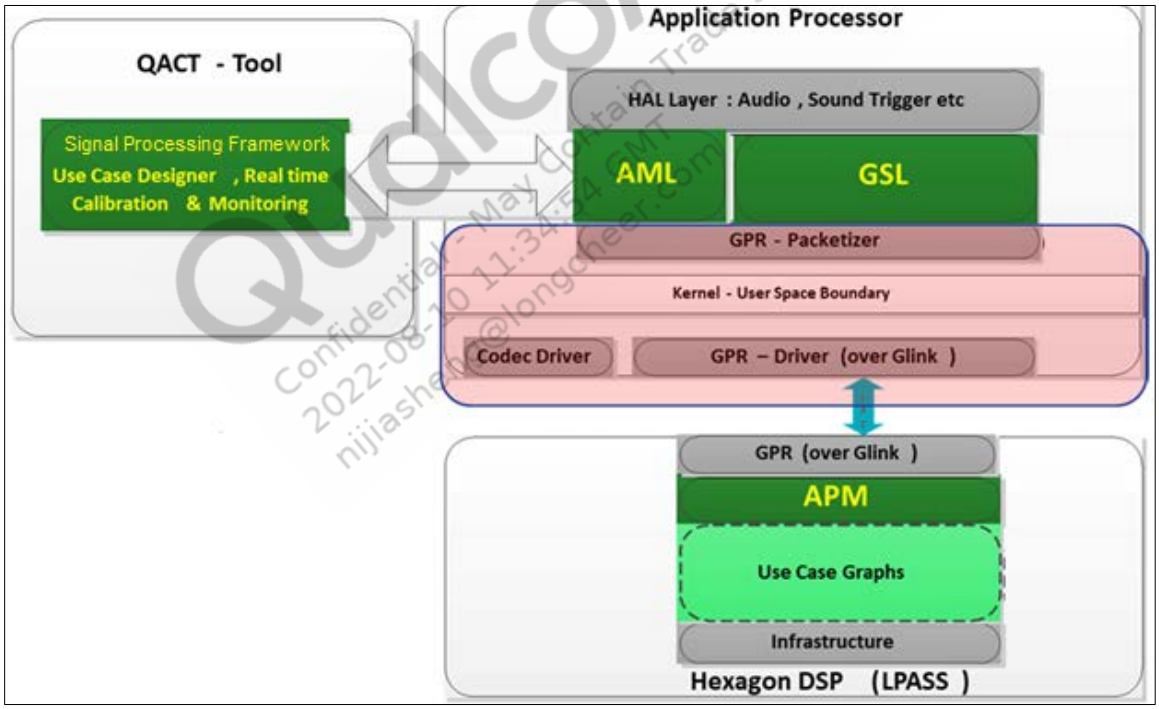
2 Overview

While realizing audio use cases, the software architecture layers residing in the application processor send a series of commands to QTI's signal processing framework (SPF) in the form of generic packet router (GPR) packets. The packets contain information that is useful for constructing audio graphs and processing the audio signal that is required for active use cases. This packet transport service that carries commands back and forth between multiple processors is referred to as GPR.

The following diagram shows GPR communication between processors using APPS an ADSP as an example.



The GPR packet sniffer tool deconstructs the GPR packets being sent to the signal processing framework in regular audio use cases. The tool can:

- Capture and parse GPR packets
- Visualize the data in tree and sequence views
- Be used to inspect and debug the packets being exchanged by the processing modules (for example, APPS and ADSP) to identify the reason why a failure occurred
- Show what commands have been sent and received by decoding the binary files to create a visual representation of the data

2.1 Tool inputs for parsing GPR packets

2.1.1 Configuration file

The config file provides the basis for what information is present in the bin file and how the tool can read it.

The config file is provided along with the software. Packet command, parameter, property, header, and module structures are translated to XML from the various headers involved in software architecture modules. The config.xml file will be updated as new commands and modules are introduced in the future.

The following is a sample config.xml file:

```
<COMMAND id="APM_CMD_REGISTER_CFG" hex_value="0x01001008">
  <Member type="header" name="apm_cmd_header_t"></Member>
  <List type="param" name="param" size="variable">
    <Member name="mod_instance_id"  size="4" display="hex"></Member>
    <Member name="param_id"  size="4" display="hex"></Member>
    <Member name="param_size"  size="4" display="decimal"></Member>
    <Member name="error_code"  size="4" display="hex"></Member>
    <Member name="payload" size="variable"></Member>
  </List>
</COMMAND>
<COMMAND id="APM_CMD_DEREGISTER_CFG" hex_value="0x01001009">
  <Member type="header" name="apm_cmd_header_t"></Member>
  <List type="param" name="param" size="variable">
    <Member name="mod_instance_id"  size="4" display="hex"></Member>
    <Member name="param_id"  size="4" display="hex"></Member>
    <Member name="param_size"  size="4" display="decimal"></Member>
    <Member name="error_code"  size="4" display="hex"></Member>
    <Member name="payload" size="variable"></Member>
  </List>
</COMMAND>
```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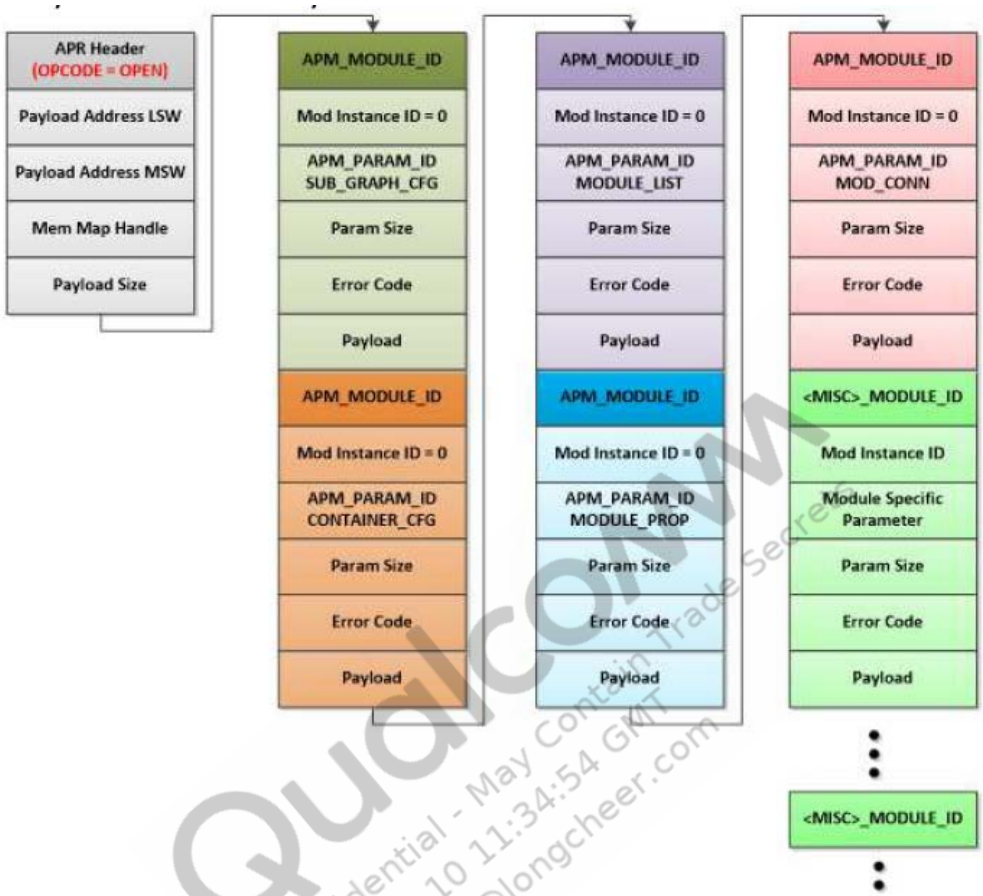
The following figure shows the expected packet structure.



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2.1.2 Bin files

The graph service layer (GSL) running on the application processor is where packets are generated and sent whenever an audio use case runs. The GSL dumps the binary file on the target whenever an audio use case is realized/run. The bin file:

- Can be pulled from the device and loaded to the tool for inspection
- Is automatically generated by default when use cases are run and is saved in the path vendor/data/out
- Is converted to a visual form once the tool parses the info correctly

GSL output of logged packets contains data in this order: total size, metadata size, metadata, GPR header, and payload. The following is a sample of a bin file.

```
73 65 6E 64 5F 70 6B 74  gsl_log_send_pkt
10 20 00 00 60 50 00 00  X........ ..`P..
01 00 00 00 00 00 00 00  ..... ..........
00 00 00 00 00 00 00 00  ................
0C 10 00 08 28 00 00 00  (...w@......(...
01 10 00 09 0C 00 00 00  ................
22 56 00 00 67 73 6C 5F  ........"V..gsl_
5F 70 6B 74 60 00 00 00  log_send_pkt`...
```


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2.1.3 QACT™ workspace file (optional)

If a workspace file is loaded, parameter names are visualized in the tool and the parameter payload is converted to an annotated format similar to QACT generic view. The parameter payload can then be opened by passing the workspace file.

The following is a sample workspace file.

```
<Key id="0xA1000000" name="Stream">
  <Index val="0xA1000001" name="PCM_LL_Playback" />
  <Index val="0xA1000002" name="PCM_Record" />
  <Index val="0xA1000003" name="PCM_Loopback" />
  <Index val="0xA1000004" name="Voice_UI" />
  <Index val="0xA1000005" name="Voip_Rx" />
  <Index val="0xA1000006" name="Voip_Tx" />
  <Index val="0xA1000007" name="Voice_UI_EC_Ref_Path" />
  <Index val="0xA1000008" name="Voip_Tx_EC_Ref_Path" />
  <Index val="0xA1000009" name="PCM_Record_EC_Ref_Path" />
  <Index val="0xA100000A" name="Compress_Offload_Playback" />
</Key>
<Key id="0xAE000000" name="MediaFmtID">
  <Index val="0xAE000001" name="PCM" />
  <Index val="0xAE000002" name="AAC" />
  <Index val="0xAE000003" name="ALAC" />
  <Index val="0xAE000004" name="APE" />
  <Index val="0xAE000005" name="WMAPRO" />
  <Index val="0xAE000006" name="WMASTD" />
  <Index val="0xAE000007" name="FLAC" />
  <Index val="0xAE000008" name="VORBIS" />
  <Index val="0xAE000009" name="MP3" />
  <Index val="0xAE00000A" name="AMRWBPLUS" />
</Key>
```

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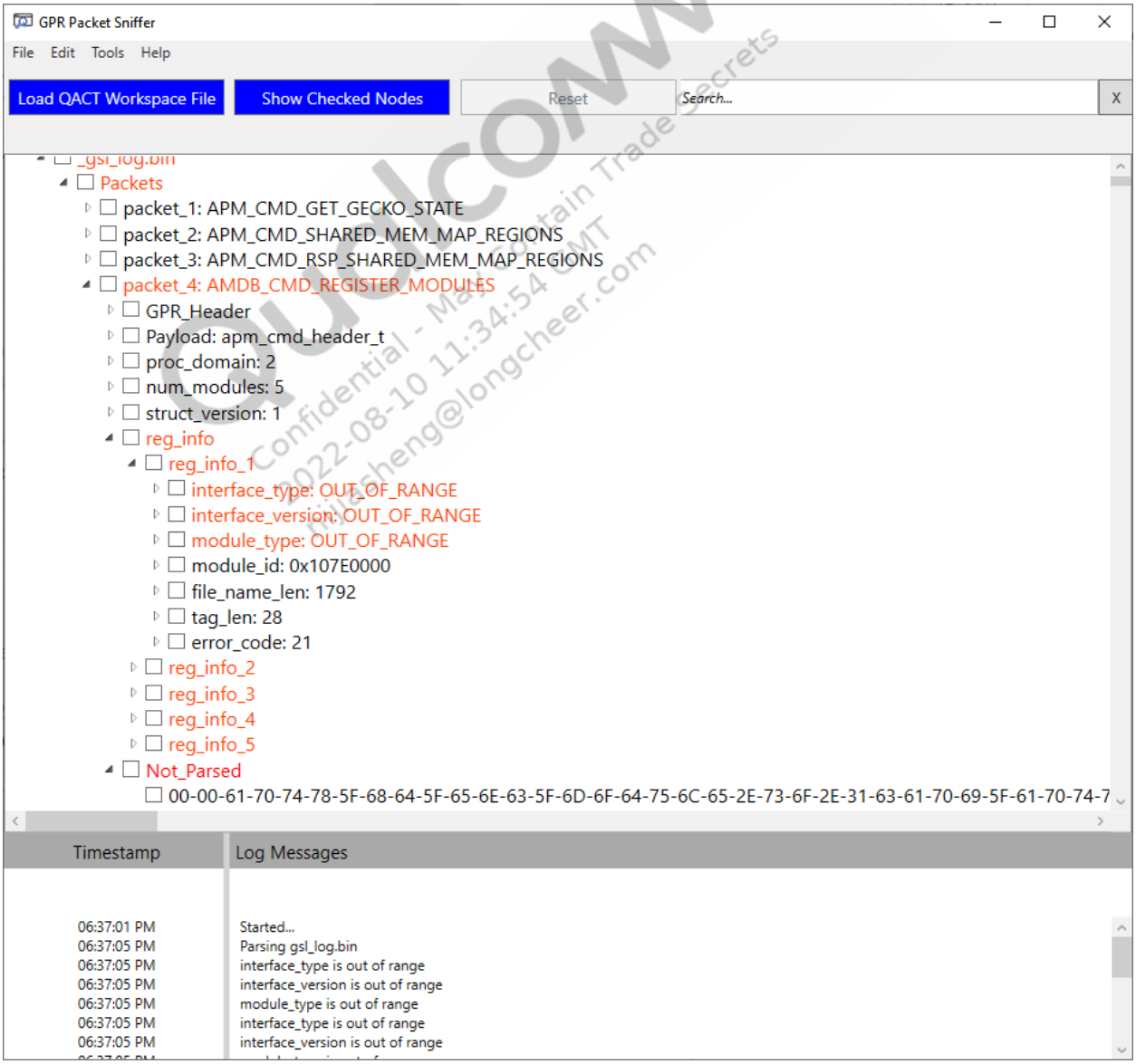
2.2 Error detection

Errors present in the parsed packets or modules are highlighted in red. There are two types of errors:

- Not_parsed – The data could not be parsed by the tool and the number of bytes that could not be parsed is shown. This could occur if, for example, the sent packet definition is not found in the config.xml that is loaded at the time parsing. To solve this error, update the xml if the structure definition is known and re-parse the file if necessary.
- Genuine error – The data is parsed properly and there is an error in the payload. The reason of the error will be specified at the highlighted sub-node.

The following figure shows packet parsing errors detected by the GPR sniffer tool.



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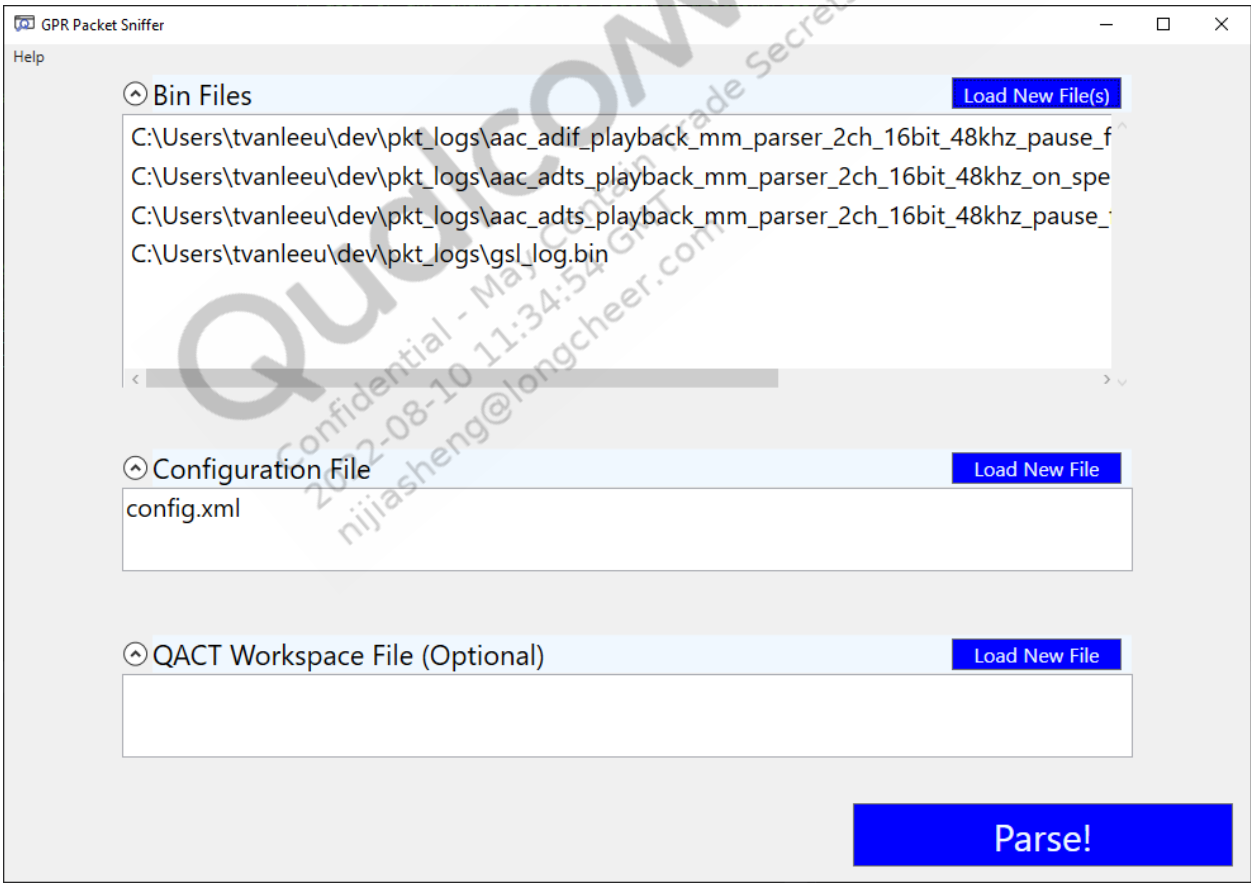
3 User interface

3.1 Main menu

The main menu is the first screen to appear when the tool is opened. Click the respective buttons to select bin files and the optional QACT workspace file. Config.xml loads from the install directory by default, but a custom file can be loaded. The binary files can be downloaded from the target after the active use case is run.

After loading these files, click **Parse**.



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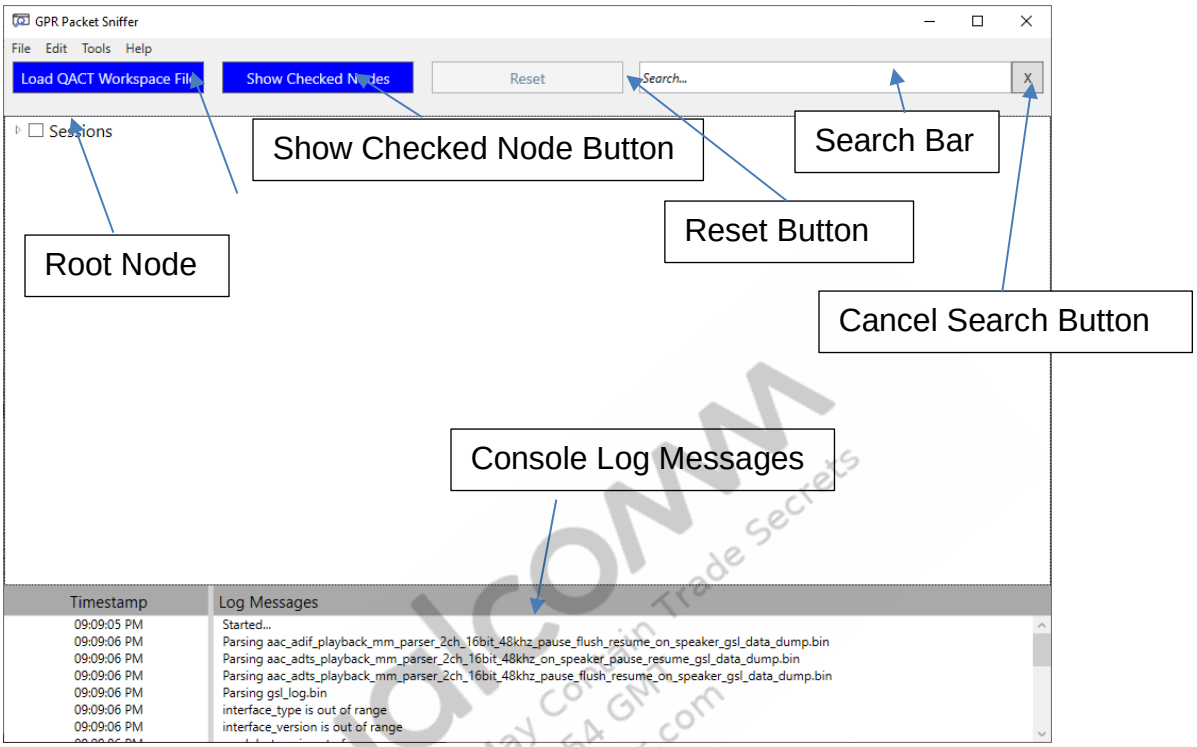
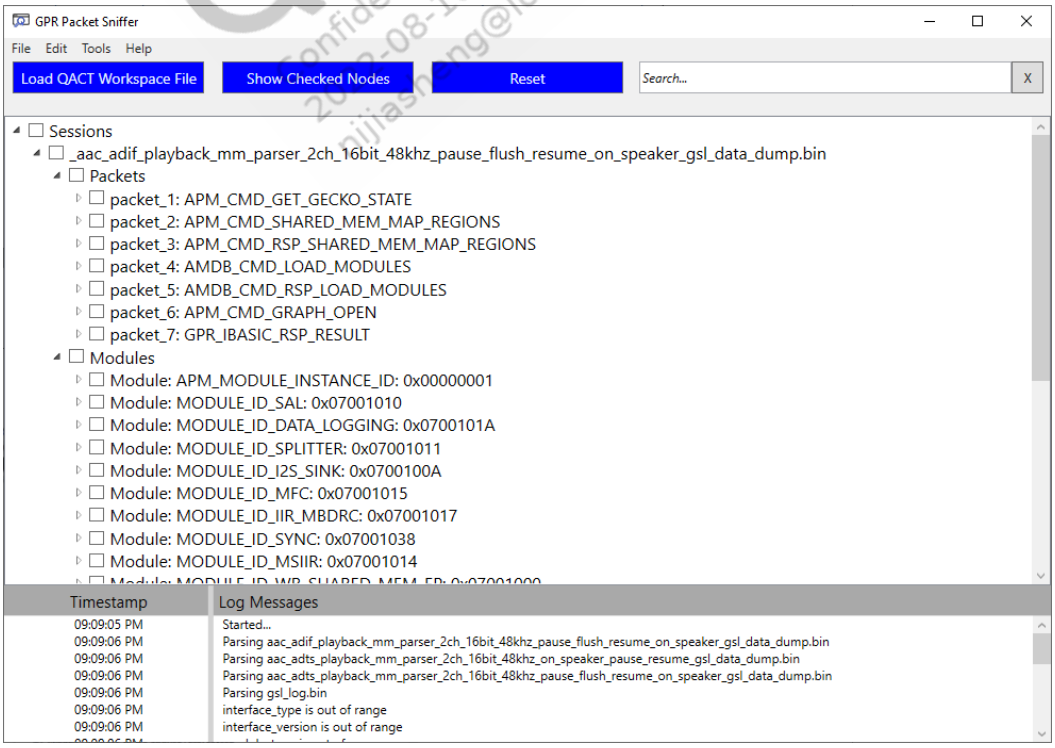
3.2 Tree view



3.3 Sample bin file structure



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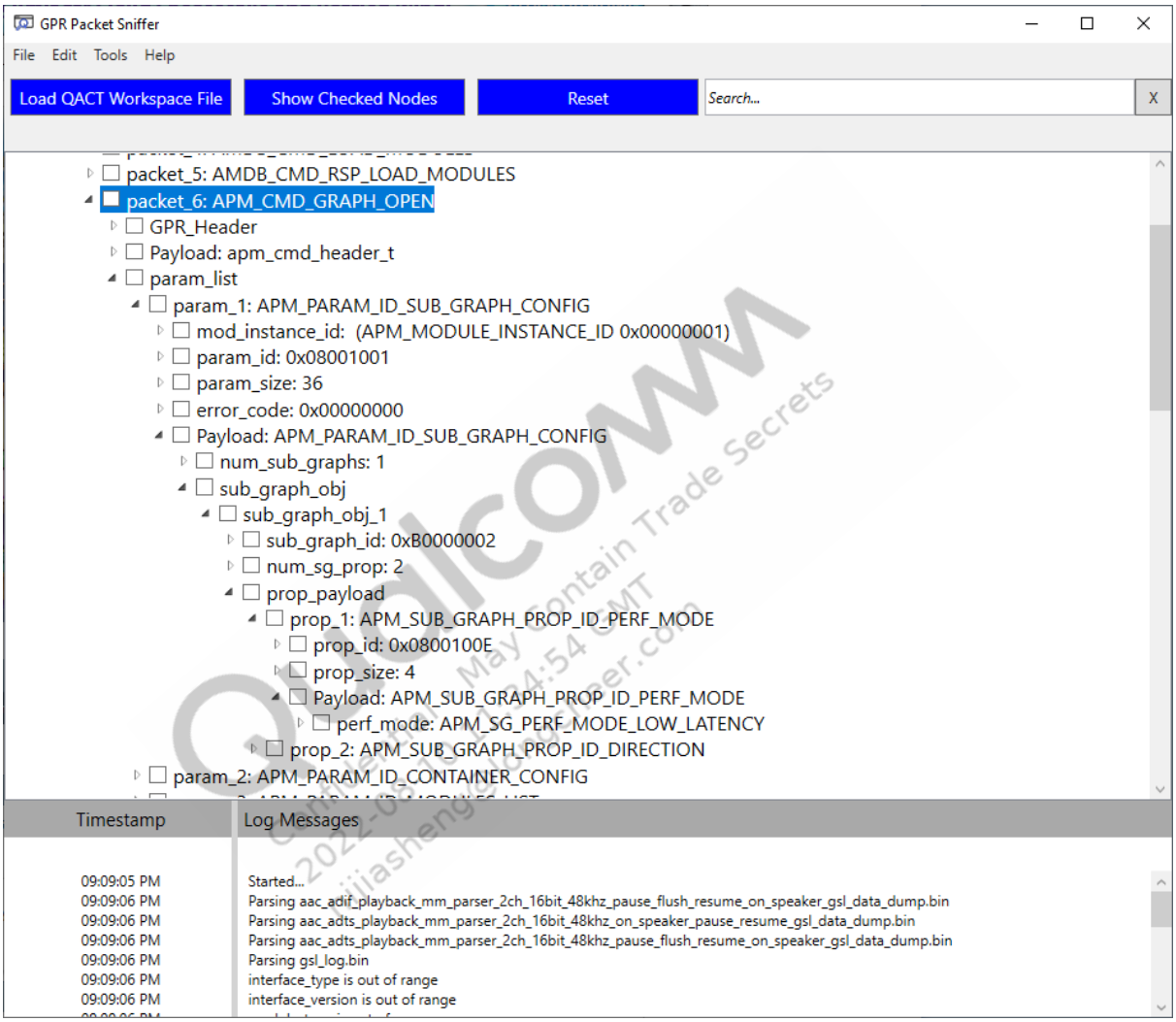
3.4 Sample packet structure

Packets can be checked to see what exact data is sent during the use-case.



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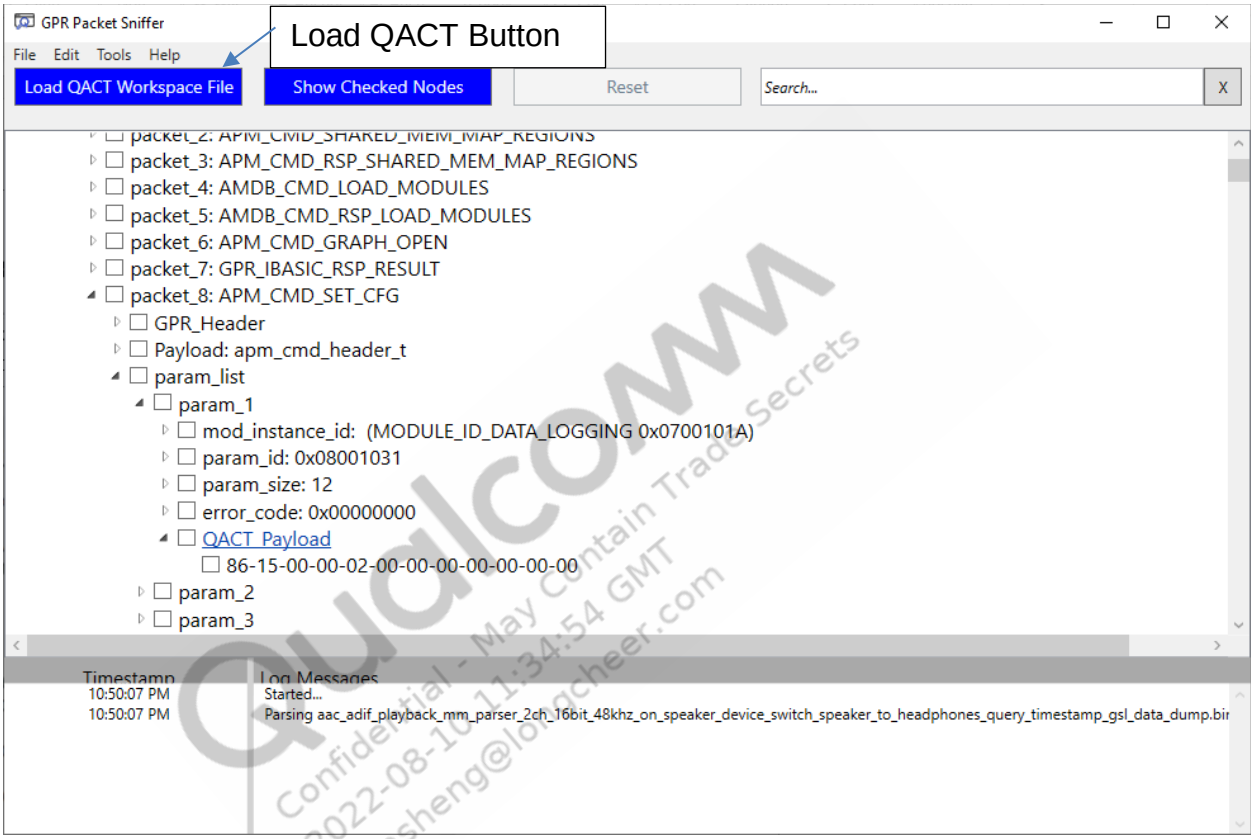
3.5 Module-based commands without loading workspace file

If a QACT workspace file is not loaded, the payload for module-based commands such as APM_CMD_SET_CFG is shown in its hexadecimal representation under QACT_Payload.



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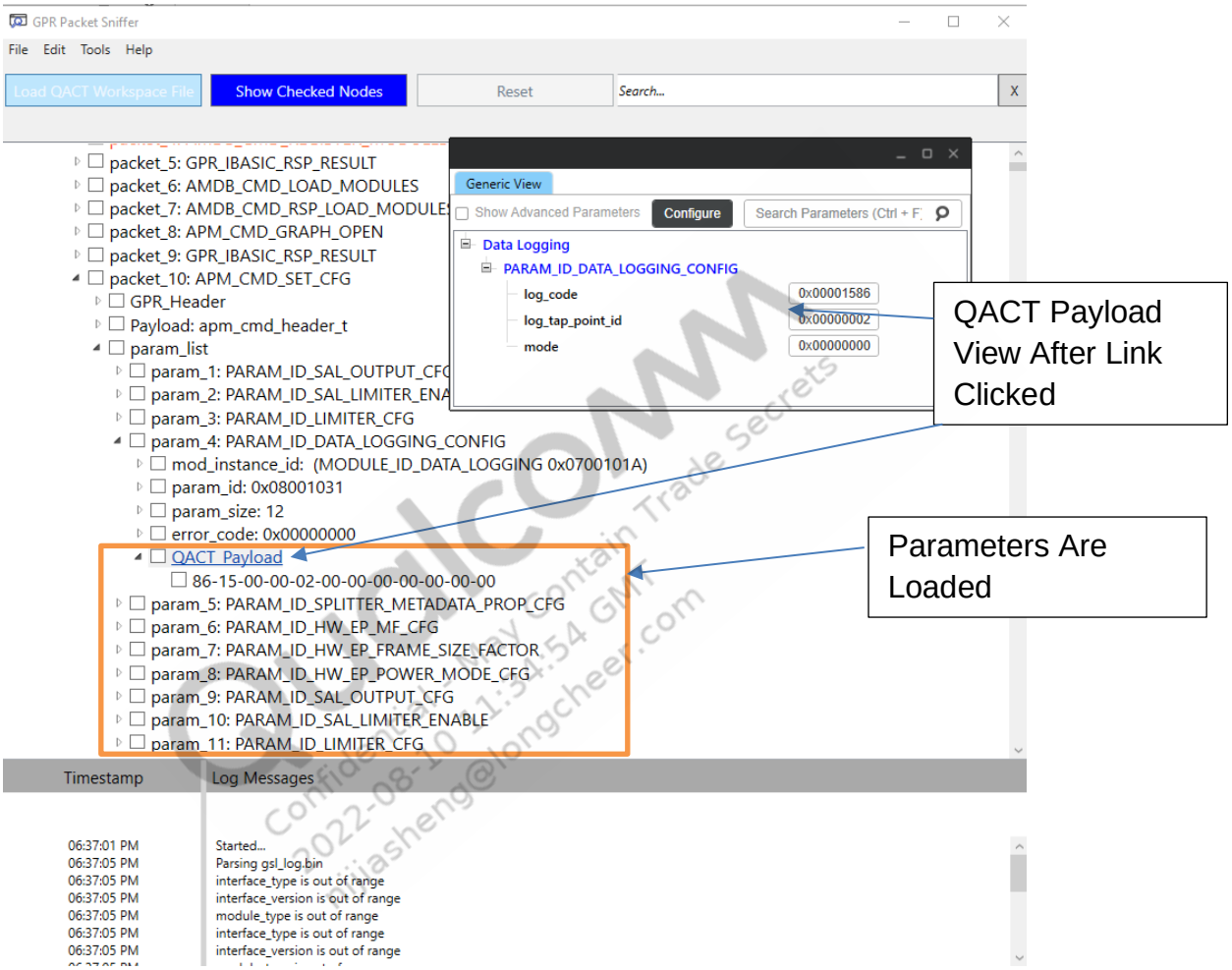
3.6 Module-based commands after loading QACT

If a workspace file is loaded, the parameters are parsed and QACT_Payload becomes clickable. Click it to display the payload.



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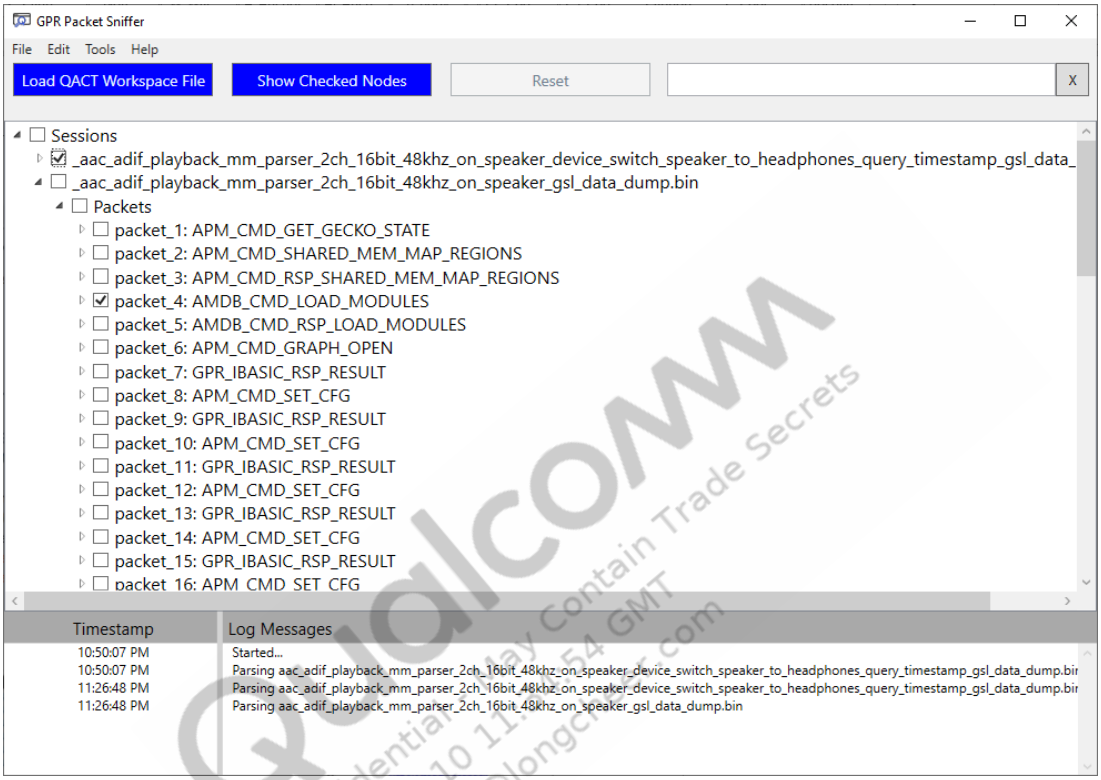
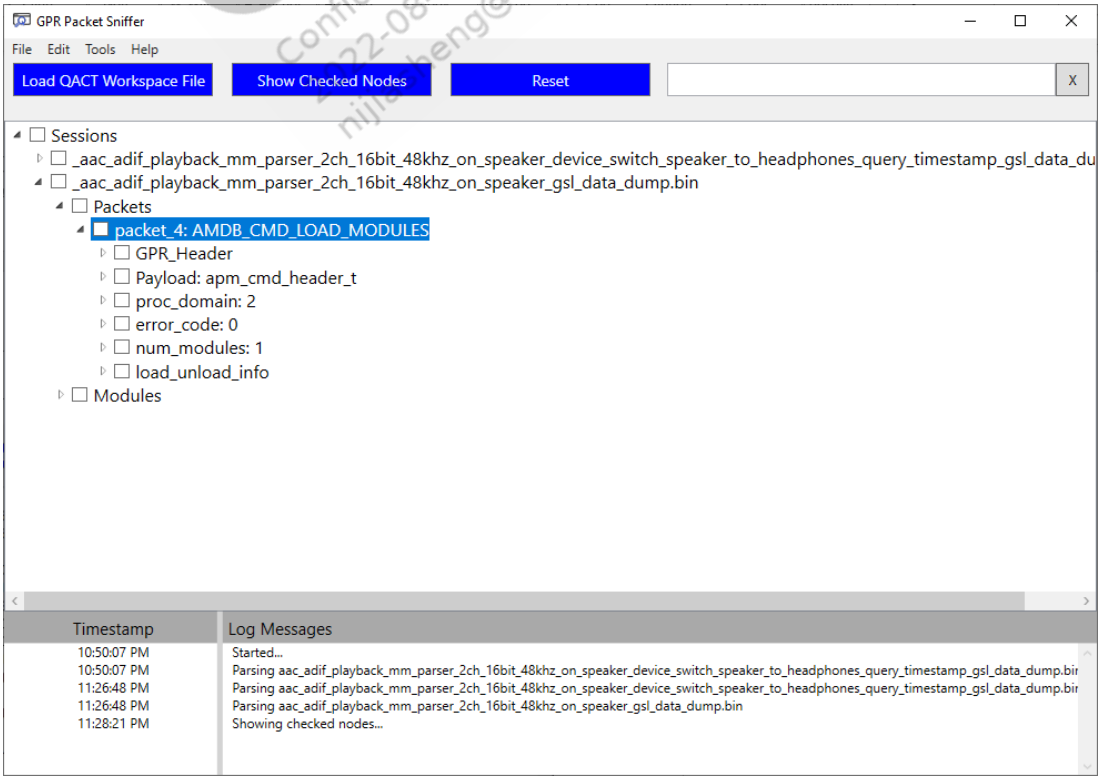
3.7 Inspect nodes

Select a node to inspect and click **Show Only Checked Nodes** to hide the rest of the packets and bin files in the UI.





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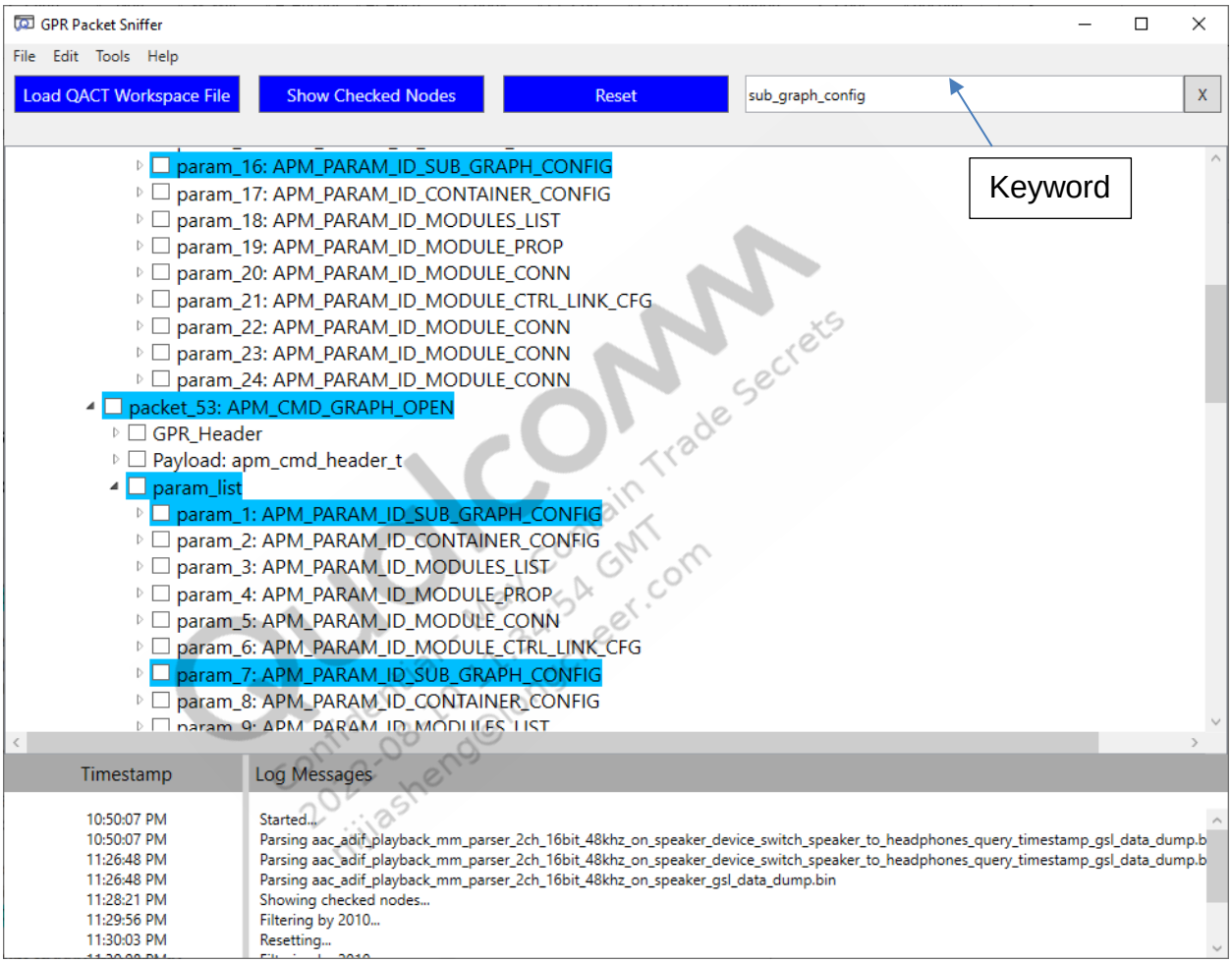
3.8 Search

Type search word(s) into the search box and press **Enter**. All instances of the words are highlighted in blue. If the word is not found, a message is printed in the console. To clear the current search, click **X**.



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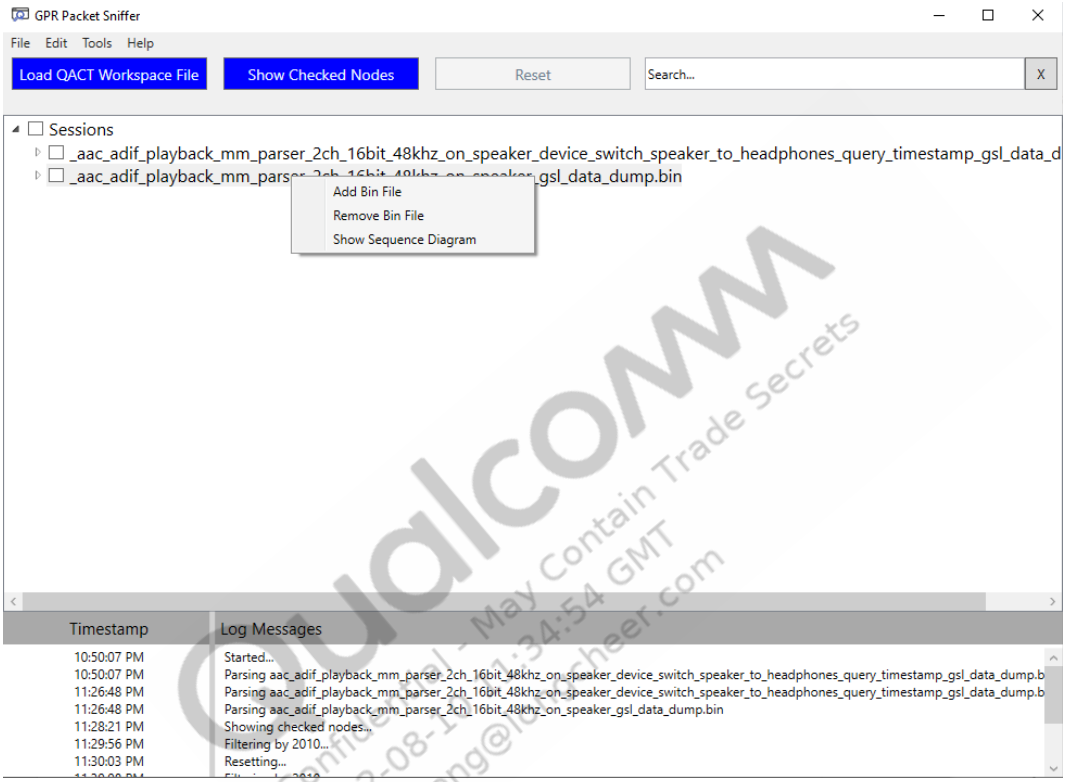
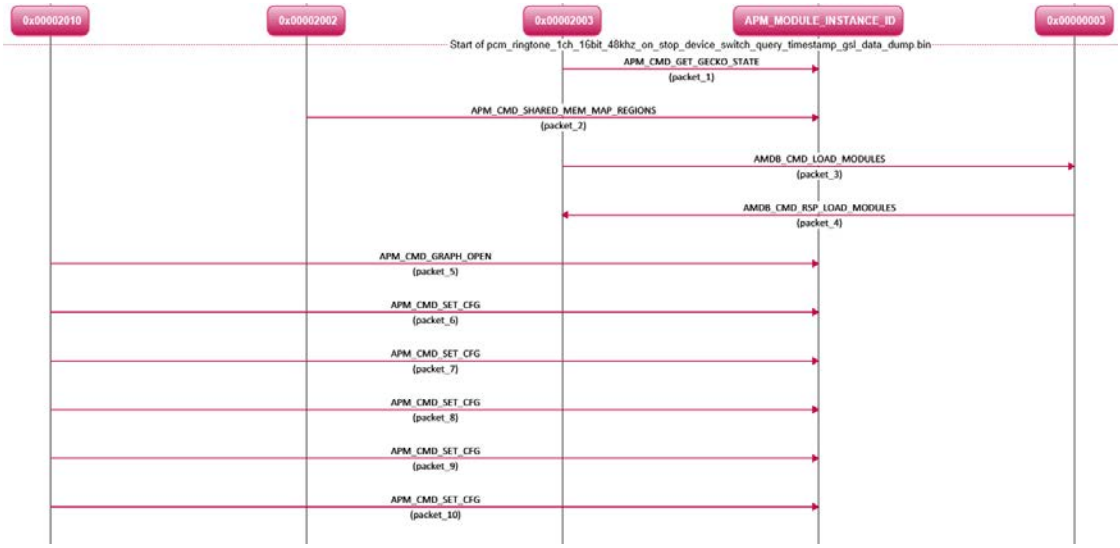
3.9 Additional options for viewing/parsing

Right-click anywhere on the form other than the console messages panel to access a context menu with options to modify the UI. When a bin file is selected, the Show sequence diagram and Remove bin file from parsing options are enabled.



The following diagram is shown if **Show Sequence Diagram** is clicked. The diagram shows the commands implemented in timeline view. It is shown as a .pdf and can be stored offline. To generate sequence diagrams, the open source tool MSC Generator version 6.2 or earlier must be installed and msc-gen.exe must be in the system path.



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3.10 Gathering log files

To enable logs at runtime on an android target, set the android property vendor.audio.gpr.enable.packet.logs to 1:

```
adb shell setprop vendor.audio.gpr.enable.packet.logs 1
```

To disable logs, set that property to 0:

```
adb shell setprop vendor.audio.gpr.enable.packet.logs 0
```

This property can be changed without restarting the device.

The log files are by default generated at /data/vendor/audio/gsl_log.bin on the target device. Copy the file to the host machine to save it.

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A References

A.1 Acronyms and terms

Acronym or term	Definition
GPR	Generic Packet Router
GSL	Graph Service Layer

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